Meeting on STRETCH & Super Stretch Plans
Poughkeepsie
Oct 27, 1960

Buchholtz - chairman	E. Adams	C. Kraus
Campbell	F. Brooks	EW Pugh
Hoffler	H. Kolsky	N. Klein
Merwin	S. Dunwell	D. Crawford
Cypser	J. Ihnen	G. Amdahl,
	R. Henle	
Seiner	Walsh	

Purpose: To Recommend what Corporation should do in area of large scale computer. -- advice to mgmt.

Report on Super Stretch: (Seiner)

- waiting for reply from AEC -- now before commission -- will hear 1st week of Nov.
- "100X Stretch" -- or "20X Stretch"

Research proposal to John Pasta interested in cryogenics

Looking into Cryotrons mainly for now or year from now

Now laying out 200 cryotrons: 16 1 bit words - full selection - full accessing -- address counters - registers - checking circuits

There is more possible payoff in cryo. -- Pasta is interested

Studies in contract? (1) AEC numerical math. techniques?

-not clear how many people,--

- Teller watson meeting
- Last Dec - meeting at Livermore
- proposal
- subsequent meetings

STRETCH (Dennell)

- no formal plans now - must be interpreted
- we could now build with ½ components, size, cost, etc.
  e.g. 50,000 Trs → 15,000 Trs + (?)30,000 diodes
  + better low speed channels (std interface)

- multiple computer: STRETCH + 4 7044's (MITRE)
  2 or other computer?
  - parts of problems on each --
  - simulation of peripheral systems in real time.

DSD's Plans (Brooks)

- next immediate effort will be in lower performance.
  $10,000 per month up (250,000) '80, '90, '70, '74 has been put out.
  ('70 is only modest machine)
- long range - above 90 - up to STRETCH - compatible with rest of line
  - not highest priority (63-64 announcement)
  - (not cryogenics or new tech.)

FSD Plans (Cypser)

- accepts resp for design, release prod. of new machine (Super Stretch)
  - not the design etc early phases
  - Research and/or DSD will do original work.

- Refer to Searson's letter: future work will be done on CPFF basis

new Technology... IBM will have to work

(Adams)

cost for complete desk; $100K sale price;

- need a lot more thru-put — large memory, needs to be 10× bigger,
- essential problem: shuffling data back & forth between high speed machines, computer net --
- only a factor of 2 increase in present STRETCH frame organization (factor of 5 if all zero) (70% of time is transmission time now)
- research into cryogenics must go on —

(Hale) 10× faster circuits are possible at present STRETCH circuit costs. COMPACT circuits, --- size would be smaller -- no changes in machine organization,
--- STRETCH is already fast on common instructions

(Rowell) ½ cost of STRETCH is already in memory.

100× machine (needs 10 or 100× as much memory)

intermediate storage -- need much more & high speed

ARPA possible customer?

(Venle) Circuits: DSD-Components -- look at computer needs. -- will put less on high speed area.

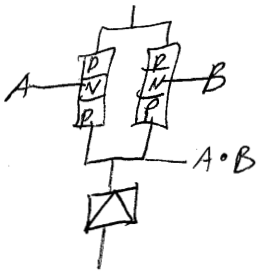
Memory -- 0,1 μs cycle

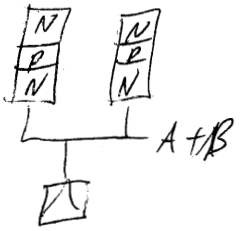
(J. Walsh)

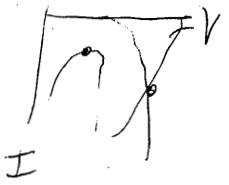
— work on faster circuits { faster diodes came in / faster transistors.

@ (5 mysec circuits) has high power levels.

started Esaki diodes - Jan '60

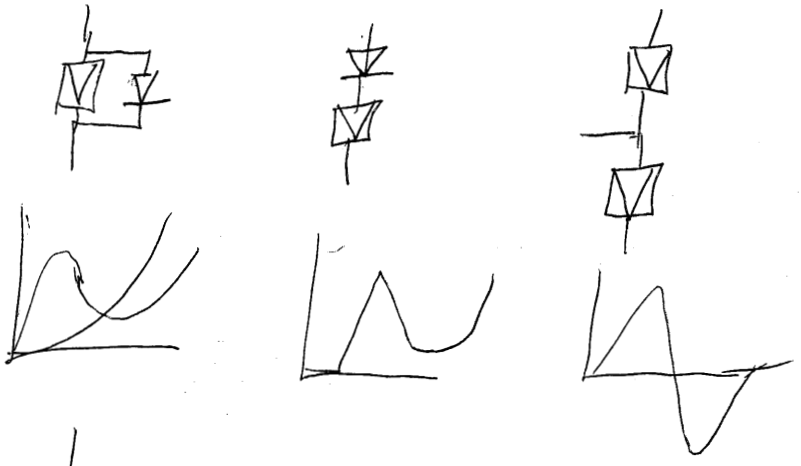


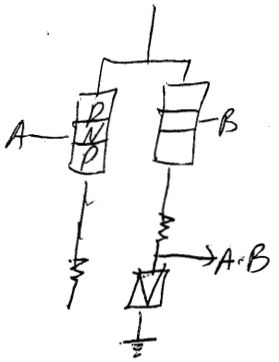




switches fast one way, slow the other

Tried to leverage Esaki diode





GA AS diodes

2 mys delays.

power goes up but is more linear

tolerances are very tight 30 mv spread

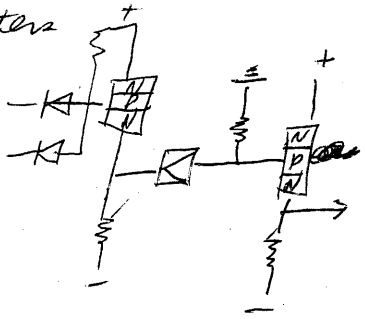
- small voltage DC circuits 5% resistors, 3% power supplies.
- 300 mv. signal swing.

→ Emitter-base drop one Tr. -- most critical factor



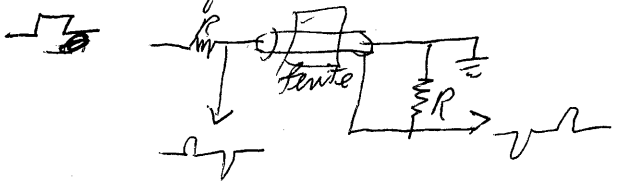
→ 30-40 milliwatts per logical block

Inverters

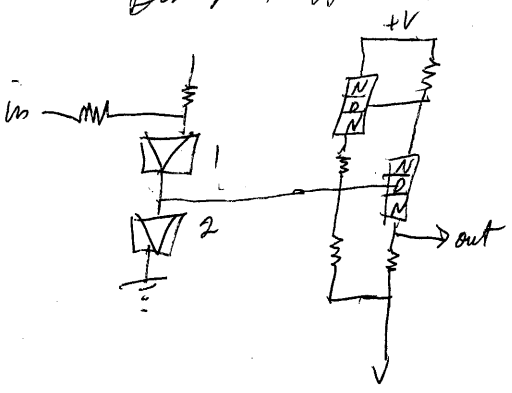


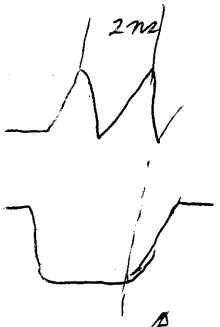
little signals are amplified - large ones are not.

Pulse Transformer: not too successful -- too much capacitance



Binary Trigger





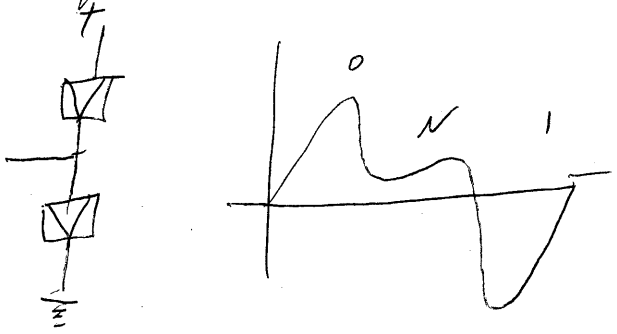
this is one shaped - not really 500 Mc

300 Mc is limit now

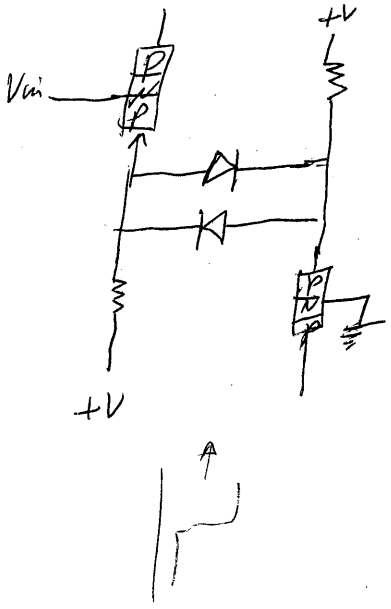
one pulse out for 2 pulses in.

Asynchronous Circuits

Required - 3 levels 1 - nothing -0



3 level current switch - not as fast



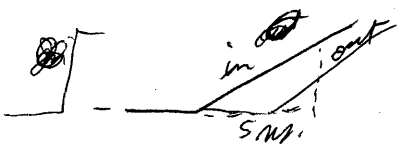
Program to Transistor analysis on 610. Esaki diodes - load lines, etc.

[expect to achieve:
100 Mc. with 1 ns delay]

RCA others?
what about ~~RCA~~ we have no advantage, ←
- low capacitance diodes are not common.
(only 14 now in existence)

packaging may be most important

5 nsec circuits:



rep rate is less
can't start to reduce input signal.



rep rate higher

(150 to 200 devices on card) ←

COMPACT order - 4×8 card - can be used - but not as easy,
- solid state strip lines - on back panel -- next extension?

Research will have a high speed machine project regardless of outside funding.

High Speed Technology is not same as large machine operation

Memories: (Crawford)

- restarted advanced development last Dec. (7 engrs + technicians now)

- partial switching techniques was looked into — 2 cores per bit.
(plenty of different switching -- which was apt)

Drive circuits: { 1 to 2 amps load voltage 30-35 v
have seen { rise times 8 ns

- driving toward 100 ns full cycle
25 ns switching time

Plans; 72 bit/word 1000 words - next summer - plan to use double selection.

{ will build 16 word crossection this year ←
{ logic & 5 ns circuits will be used for 16 wd

16 K. 1/4 µs mem now being worked on
→ - relaxation effect -- serious limit after pulse to settle down
e.g. 25 ns pulse don't reach disturbed threshold for 2 or 3 µs } other materials are ok down to 0.1 µs
at 1/4 µs down 30-40%

- The above is limit for rep rate on cores

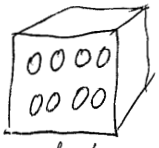
100 ns cycle) 10 Mc are questionable now for cores,

{40-50 mvolts, signals
-gain needed 10

- cores out of thin metallic films (cylinder glass tubing)
- better below 10-15 ns, - less drive currents & full switching

hope for future

{signal is variable depending on width



20 mil tubes in plastic
- print, printed wiring

(maybe 1000 wd memory will be these)

~~Tunnel Diode~~ memory
- flat thin films

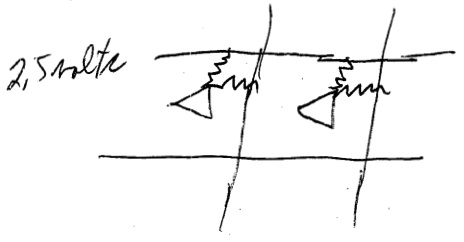
{3 or 4 mvolts for flat thin films

Esaki Diode memory

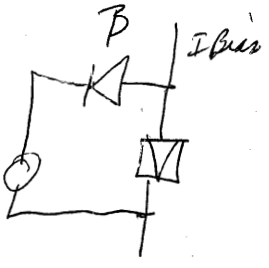
home built: 16 words 32 bit/wd.

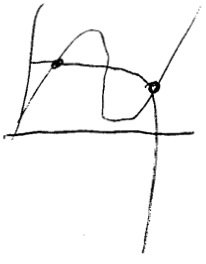
150 ns cycle -- can be cut in several ways,

"parallel-parallel" selection



amt of current in a cell depends on state -- variable load on driver (serious problem)





get good pulse for 1, very little for 0,

may go to 50 ns

Read only memory -- will look into next year -- low cost, - study effort,

flat thin films (Pugh)

competition is up or un

Sperry Rand,

1000 word 50ns, memory now being decreased,
low level outputs, (maybe 30ns) – so really 100ns

100ns within 1 year ← target of French lab.

50 x 50 bit
wds.

Higher – multilayered films ? plated toroids

1.5 kMc Transistors can be made at not too much
additional cost – very small – in 1 year,
– cheaper packaging is important, (basic cost 1¢ ea – package much more)
silicon – passivating circuits ? – glass – cut cost of headers,
"surface pasification"

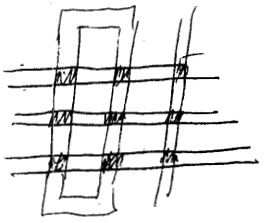
NPN Mesa Transistor – fastest in quantity now.

Cryogenics (Ittner)

ADL plans to build small assoc mem.
STL ?

- costwise would be quite favorable -- if it can be assembled in large quantities.

Model for circuits -- optimum, assume { $\rho_s = .007$; d = insulation thickness 5000 A°



F_i = fan in of 2 (4 controls) 4 gates

F_o = fan out of 4 (8 controls) 4 1st 2 gates
4 2nd controls 3 gates

	2L/R for loop.	
1. unbiased crossed film (4 to 1 crossing ratio.)	120 ns	12 cryotron/loop
2. biased crossed films	10 ns	
3. In line cryotrons	5 ns	

Perhaps gain factor of 2 with thinner insulation - Time scale? order
" " 2.5 with alloys - by end of year '61,

So may get to 1 or 2 ns

Inductance of line = $L \sim (d+\lambda)\left(\frac{l}{w}+1\right)$

λ = penetration depth into film { 500 A lead { 1500 A alloys, tin

assumes 1 width between cryotrons.

Resistance of line = $R \sim \rho_s \frac{l}{w}$

l = length of cryotron
w = width

$$\frac{L}{R} \sim \frac{d+\lambda}{\rho_s}\left(1+\frac{w}{l}\right)$$

$$H = 300\left(1-\frac{T}{T_c}^2\right)$$

Temp.

Power dissipation: per switch $= \frac{1}{2}Li^2$

constant switching $= \frac{\frac{1}{2}Li^2}{2\tau/R} = \frac{1}{4}i^2R = \frac{1}{2}i^2\rho_s\frac{l}{w}$

↑ these can be changed

d insulation thickness	ρ_s needed for 1 ns
500°A	.014
1000	.017
2000	.023
5000	.041

.007 ohms/sq is value for TiN being used now.

Field $H = \frac{4\pi I}{w}$ ← .006" width

6–8 oersteds nominal value

so $I \approx 100$ ma

$i^2 \approx 10^{-2}$ $\frac{l}{w} \approx 10^2$

Power $\doteq \frac{\rho_s}{4} \equiv$ 5 milliwatts / cryotron (100 milliamp switching ½ ns)

or 200 cryotrons / watt 1 watt ≈ present cryostats

if one gate in each loop were being used all the time,

$\frac{2\times10^2}{2\times10^5} = .001$ efficiency,

1 watt in the watt need 3 horsepower motor

3 watts — 5 hp.

- now considering 4"×4" substrates.

→ by next June '61 will know how well it works

(Haus) 40 bit mem has been built at Kingston

135. $2'' \times 1\frac{5}{8}''$ 19 evaporations tolerance ±1% (±1.5 mils) ~$\frac{1}{8}$ part

→ will need ¼ mil registration eventually,

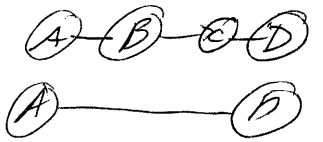
3μs read cycle
4μs read-write cycle
} present speeds.

(Lewin) Problems: (1) coupling high speed internal with slow external -- variable speeds of cryotrons may be useful

6 mil w = l

40 to 50 bits/sq in for regular memory array
~half for other circuitry,

- linear distance between exit gate & gate being driven counts:



} about same speed

circuit speed ~ perimeter of the plate

So machine would have to be organized into little units,

interplane connections? -- 4NS leads have been done, -- requires considerable power,

(2) cross over problem -- connecting planar elements together

- no. of layers is serious problem -- {20 or more layers of circuits

- cryotrons seem to be more promising for small memories ? than for large memories, - switching circuitry are same as mem.
- larger memories will be slow ~10 μs, (200,000 words layout 4 μs)

speed depends on shape of circuits not size ??

1000 per square inch - present core densities } if this is much
i.e. 30 per linear inch } better than cryotrons now,

1.2×10^6 bits 5"x5"x24" ← core memory now,

External Storage:

- new tapes - plated tapes are coming along,

TRACTOR: 3000 bits/inch 10% phase shift

plated tape: 4500 bits/inch no detectable phase shift.

- nothing is worked now

Disk: 50 micron separation being sought between disks & head

Magnetic strip -
transfer from drum to tape



Coding memory: - different length delay lines
get gain of type with zero start time

~~Cryotron~~ - continue to push high speed components & cryogenics,
- recognize that IBM will build a super machine,
- must at least match RCA
- machine organization
- problem studies } application studies on SDD
- should obsolete STRETCH

Sightseeing: 1965 ~~Bi-megacycle~~

- Rem Rand Thin film
- RCA ~~Thin film~~ tunnel diode (microwaves earlier)

IBM cryogenics -- have been pushed --
(no working hardware was shown)

RCA will no doubt get contract. --
(RemRand – memory · RCA – logic) IBM might be able get if cryogenic.

Burroughs -- a special machine

Recommendations:

1. Now is the time to get a machine study contract
2. Continue to support high speed circuits & memories devices.
3. IBM will build a 10x to 100x machine
4. Don't let make AEC & NSA machines the same agency.
5. Application studies be undertaken

IEG Oct 31,

above are bride point? Policy 100% by gov't money if DSD, except for component res. But this can be violated in specific areas, eg, files, memories – might be DSD,

But 100% funding can't hold since IBM is going to work in areas anyway -- cost sharing? when time comes – CMC will decide. will DSD give up or not?